

Group Topic: Fraud Detection

Credit Card Fraud Detection Using Machine Learning

Utilizing the 2023 Kaggle Dataset for High-Precision Financial Security
Modeling

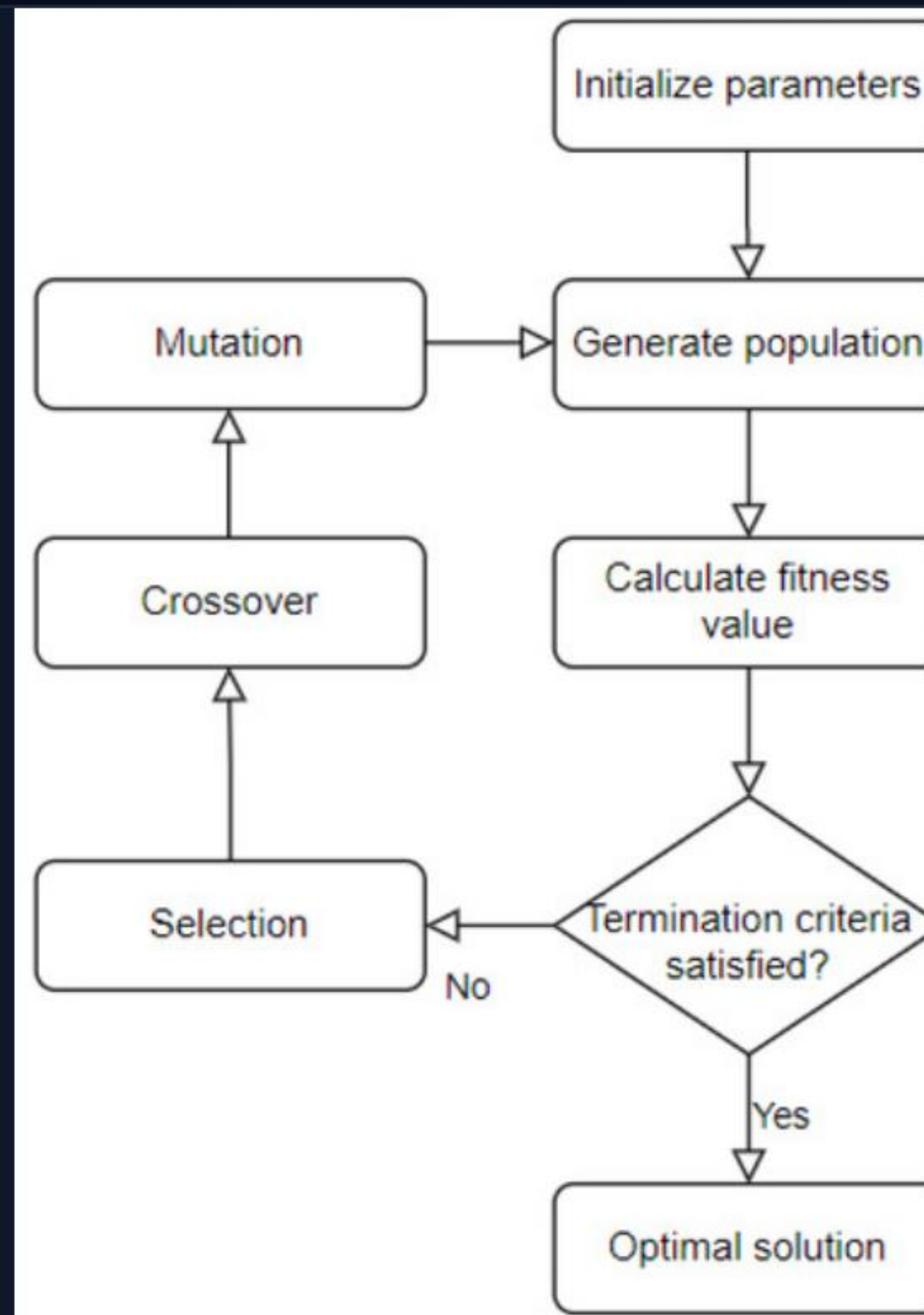
GROUP MEMBERS:

-  Nkeng Sunday
-  Kenyaupli Lucienne
-  Taiwo Comfort

DATASET SELECTION & SCHEMA

The study utilizes the **Credit Card Fraud Detection Dataset (2023)** from Kaggle, authored by *Nidula Elgiriye withana*.

-  **Numerical Features:** V1–V28 (PCA transformed)
-  **Financial Data:** Transaction 'Amount' included
-  **Target Variable:** Binary 'Class' column



WHY USE THIS DATASET?



Large Volume

Massive sample size ensures robust model training and statistically significant results.



Privacy First

Anonymized features via PCA preserve user identity while maintaining predictive power.



Balanced Classes

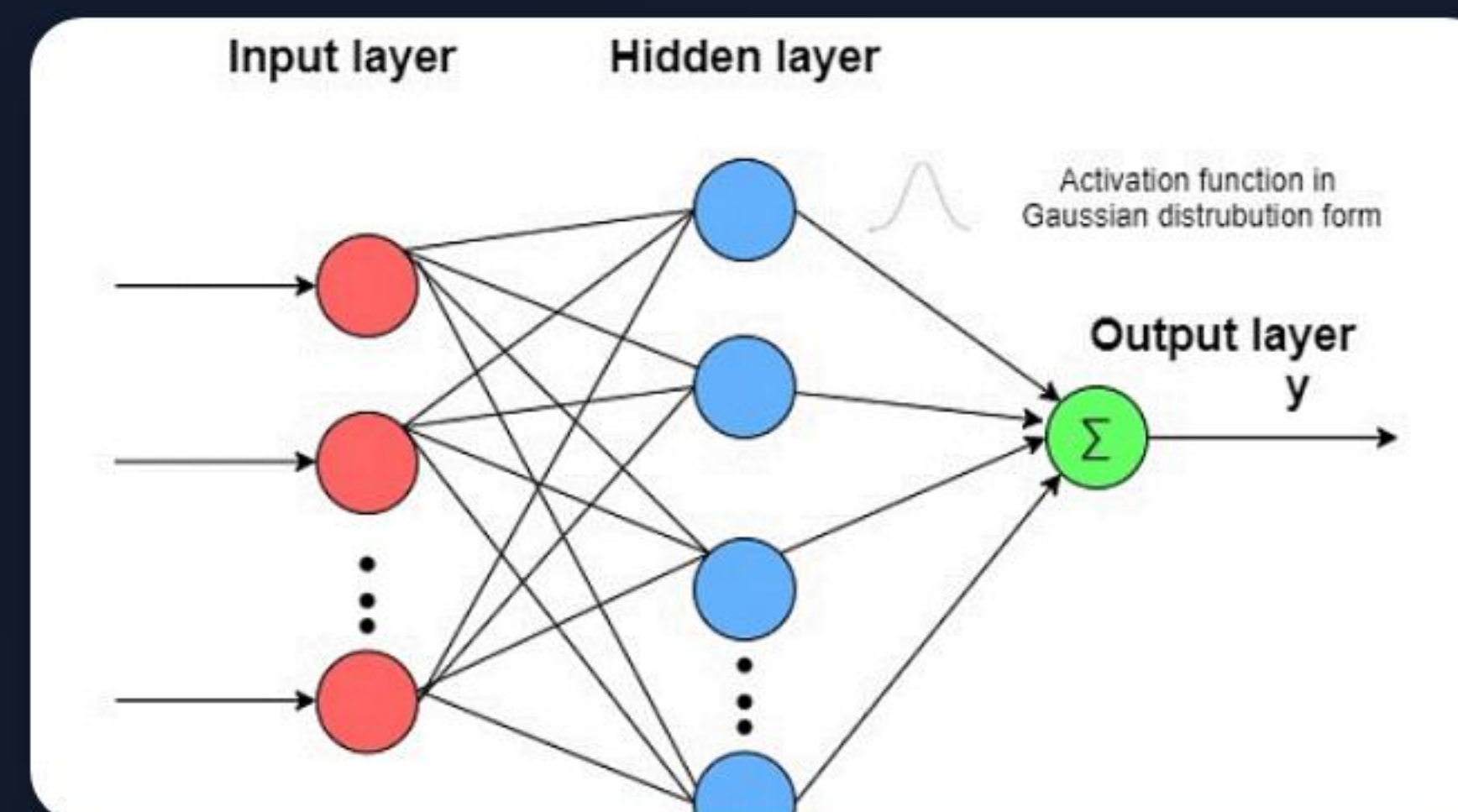
Native class balance removes the need for SMOTE or undersampling, avoiding metric bias.

THE TECHNICAL STACK



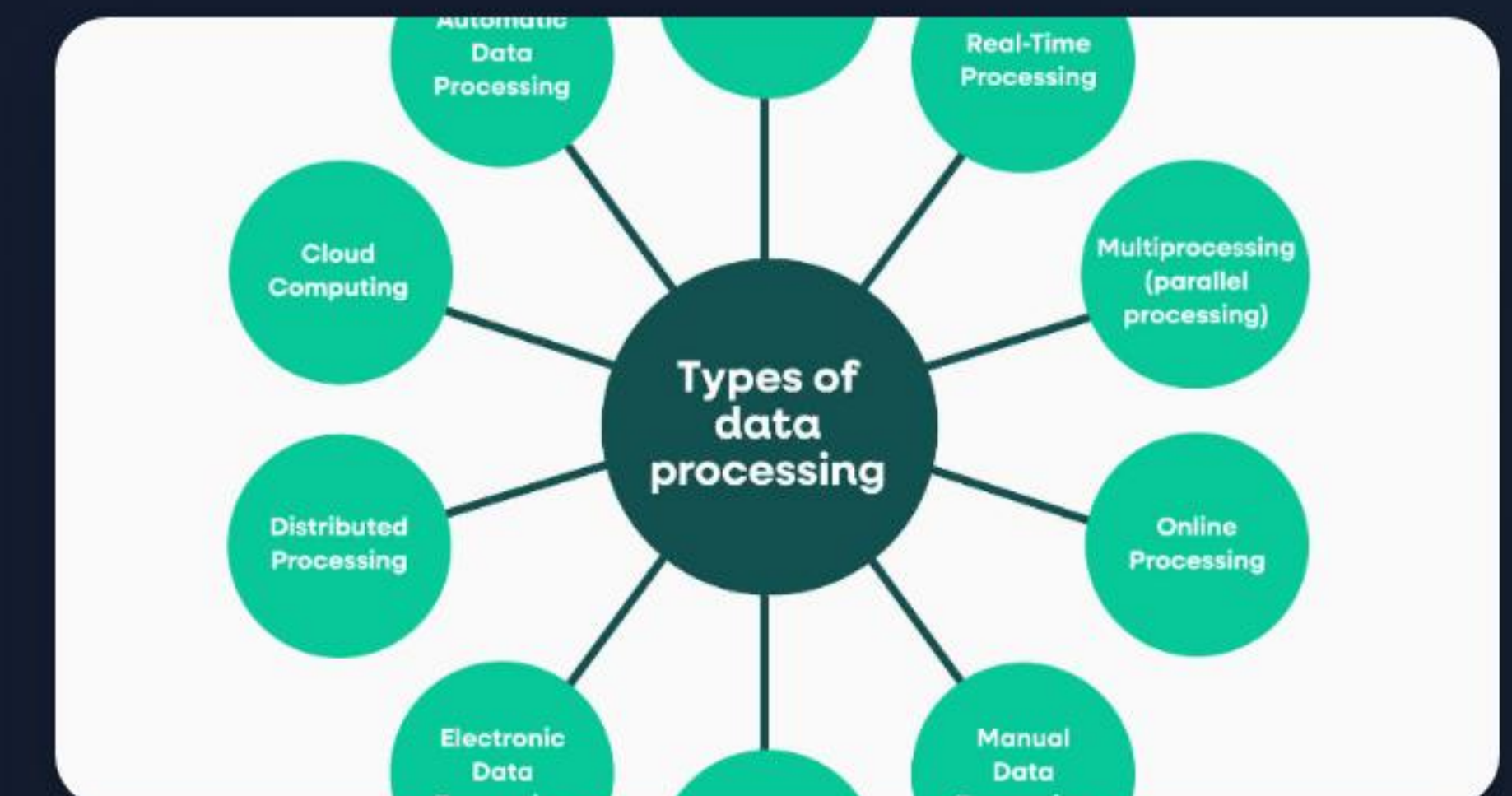
NumPy & Pandas

Core Data Handling



scikit-learn

Modeling & Evaluation



Matplotlib

Performance Visualization

EXPLORATION & CLEANING

-  `df.info()`: Verified numerical data types only.
-  `df.isnull().sum()`: Confirmed zero missing values.
-  `df.duplicated()`: Zero duplicate rows detected.

Verification Insight

The dataset is verified as "clean" and ready for modeling without the need for imputation or complex outlier handling.

CLASS DISTRIBUTION BALANCE



■ Class 1 (Fraud): 284,315

■ Class 0 (Normal): 284,315

Perfectly 50/50 split ensures accuracy and precision metrics are reliable benchmarks.

PREPROCESSING & SCALING

29

Total Features Selected

Standard Scaling Applied

The 'Amount' column exhibits a high numeric range. We utilized **StandardScaler** to normalize this feature, ensuring it doesn't dominate the PCA-transformed variables.

$$z = \frac{x - \mu}{\sigma}$$

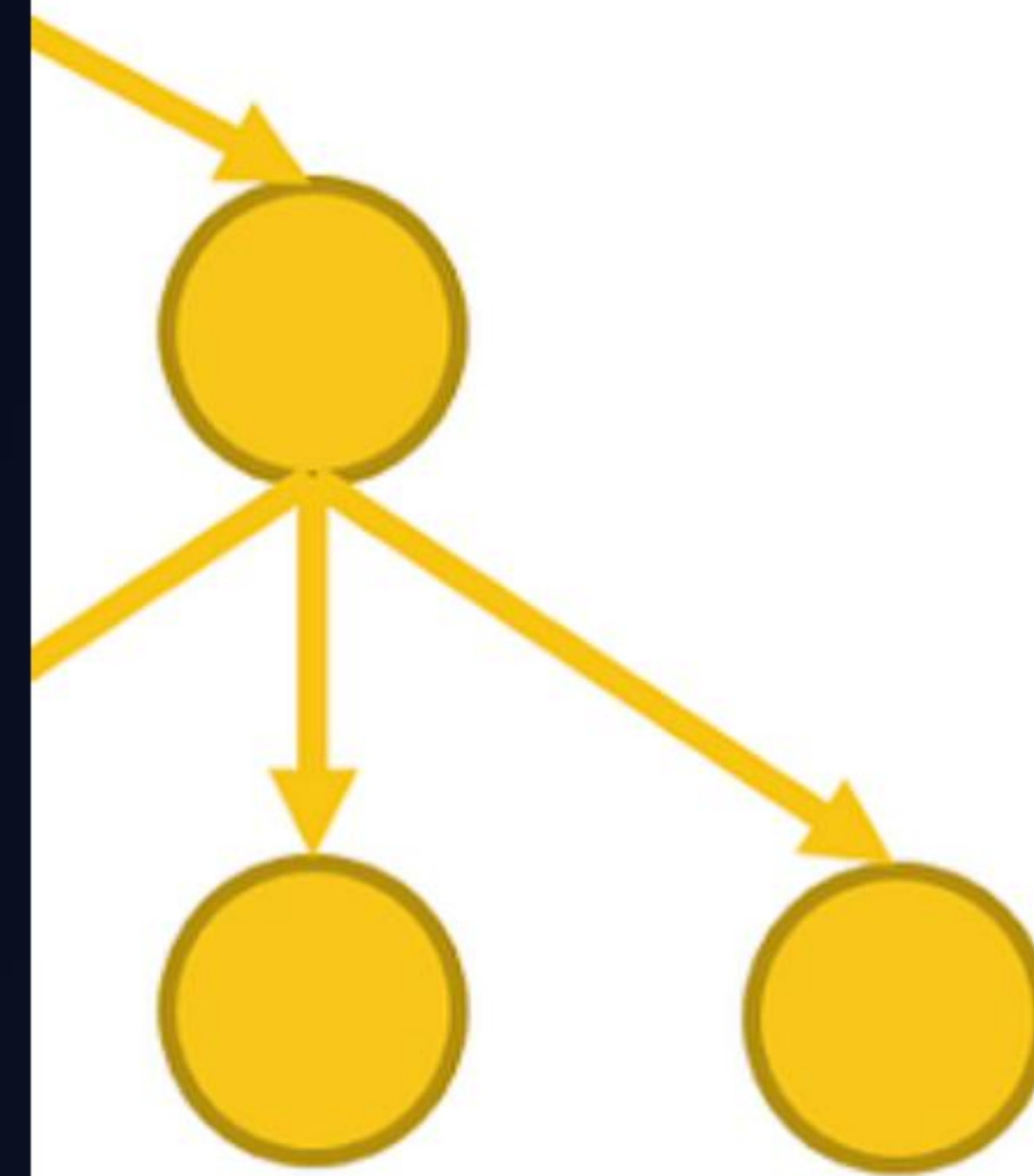
RANDOM FOREST CLASSIFIER

Why Random Forest?

Handles complex, non-linear fraud patterns effectively through ensemble decision trees.

Significantly reduces overfitting compared to individual decision trees, providing stable predictions on unseen test data.

Excellent scalability for large transaction volumes.



CONFUSION MATRIX RESULTS

CONFUSION MATRIX	PREDICTED: 0	PREDICTED: 1
Actual: 0	56,744 (TN)	6 (FP)
Actual: 1	14 (FN)	56,962 (TP)

High True Positive rate demonstrates exceptional sensitivity to fraudulent activity.

PERFORMANCE EVALUATION



The model achieved near-perfect scores across all critical fraud detection benchmarks.

STRATEGIC CONCLUSIONS

High Efficiency

Random Forest performs excellently for binary classification in fintech, effectively separating fraud from normal transactions with minimal Type II errors.

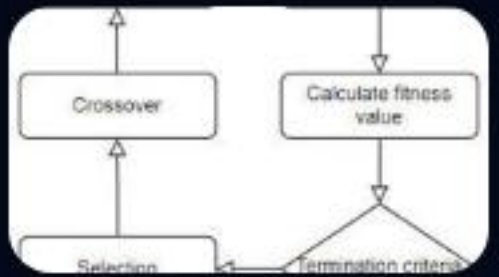
Reliable Architecture

The combination of balanced data and ensemble modeling provides a sufficient and effective approach for real-world fraud prevention systems.

Presentation Conclusion

Nkeng Sunday • Kenyaupli Lucienne • Taiwo Comfort

IMAGE SOURCES



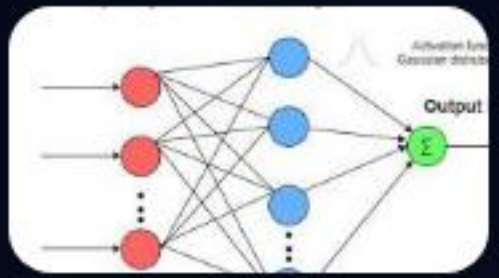
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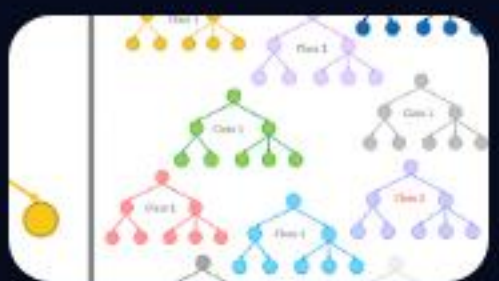
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Source: institutoalanadesa.com.br



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Source: www.fullstory.com



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Source: medium.com